EC991 Topics in Macroeconomics

Week 1 Tutorial · Your computing toolkit

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The Modern Toolkit for Computational Macroeconomics

Tool	What it is	Why it is here
Julia	The programming language	Fast loops, reads like maths, used in the heterogeneous-agent literature
VS Code	The editor	One window for files, REPL, Git and the agent; the Julia extension links them
Git and GitHub	Version control, and a host for it	Your work is never lost; you can read exactly what changed, and who changed it
Pluto	Reactive notebooks	Explore a model with no hidden state; the labs start here
opencode	An AI coding agent	Writes and edits code under your supervision; open source, works with free models

- If you already use another editor, agent or language, you may keep it; the **habits** are what matter
- Everything on this list is free for students; nothing today needs a card
- Excellent material is available: **Computational Methods for Economists** (Fernández-Villaverde)

Why learn to code when AI can do it

- A few years ago the hard part was **syntax**: which bracket, which index, why the loop overruns
- Language models have made that part cheap: ask, and you get a value function iteration
- So why a whole tutorial on tooling?

Because the barrier is lower, not gone.

- The agent **writes**; you **decide** whether what it wrote is right
- That decision needs three things: read code fluently, know what the answer should look like, test it
- In this course every number gets checked against something known: a closed form, a theorem, a limit
- An agent cannot do that for you if you do not know what to check

Programming with AI now mostly means you supervise the loop. Learn to supervise it well!

Julia: why this language?

Three reasons.

- **Speed where it matters**: a plain `for` loop over a grid is as fast as compiled code
- **Reads like the maths**: Greek letters, `u.(c)` to apply a function to a whole grid
- **The literature uses it**: much current heterogeneous-agent code is in Julia

Present value of a constant income y over $t = 0, \ldots, T$, as you would write it on paper:

```
pv(y; beta = 0.96, T = 10) = sum(beta^t * y for t in 0:T)
pv(1.0; T = 10_000)                # close to 1 / (1 - beta) = 25
```

Python.

- Wherever I can, I provide the **same code in Python**; the week 1 notebook already has a twin
- For the problem sets, **use the language you prefer**
- The ideas transfer: grids, iteration, interpolation, distributions. The syntax is the small part.

Julia: how it is organised

juliaup installs Julia and keeps it current. Avoid downloading Julia from a web page.

The REPL is the interactive prompt. One key at the start of a line switches mode:

Key	Prompt	Mode	Example
`]`	`pkg>`	packages	`add Plots`, `st`
`?`	`help?>`	documentation	`?maximum`
`;`	`shell>`	shell command	`;ls`
`\beta` then Tab	`julia>`	Unicode input	gives β

Environments. Each project records its packages in `Project.toml` and `Manifest.toml`

- `activate .` then `add Plots` or `instantiate`; `st` shows where you are
- **Read the prompt**: `(@v1.12) pkg>` is the global environment, `(ec991) pkg>` is your project
- Nearly every "package not found" error is a package added to the wrong environment

VS Code: one window for everything

- The REPL is where you **experiment**; VS Code is where you **write files**
- The **Julia extension** links the two: cursor on a line, `Shift+Enter`, the line runs in a REPL below
- Open the **folder**, not the file: the extension activates the folder's environment for you

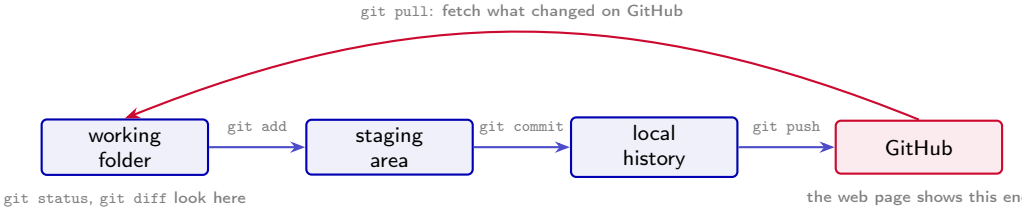
Five shortcuts for day one (Mac: Cmd for Ctrl)

Keys	What it does
`Shift+Enter`	Run the current line or selection in the REPL
`Ctrl+F5`	Run the whole file
`Ctrl+Shift+P`	Command palette: `> Julia: Start REPL`, `> Git: Clone`
`Ctrl+‘`	Integrated terminal (this is where Pluto and opencode run)
`Ctrl+Shift+G`	Source control pane: stage, commit, push without typing Git

Plots open in a pane inside the editor. Hovering over a variable shows its value once the code has run.

Git: a history of your folder

A **commit** is a snapshot with a message, an author and a time. **GitHub** hosts the history online.



Three reasons you want this.

- Your code is **never lost**, and you can always go back to a version that worked
- When an AI agent edits your files, `git diff` shows **exactly what it changed**
- Collaboration and submission run through it; so does the course repository you will clone
- Commit small and often, with a message that says what and why

Pluto notebooks

- A web page of Julia cells; **reactive**: change a cell and every dependent cell re-runs
- No hidden state, so a notebook always means what it shows; good for exploring, poor for long runs
- Packages install themselves and are recorded in the file: anyone who opens it gets the same setup

Three rules that trip up everyone the first time.

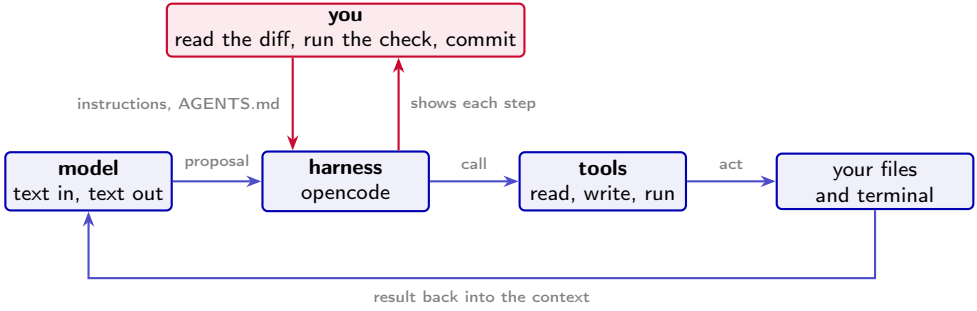
1. **One definition per cell**: to change `x`, edit its cell; do not define it again below
2. Several lines in one cell need `begin ... end` (shared) or `let ... end` (local)
3. **Open the notebook through Pluto, never as a script**

How to start it (from the VS Code terminal): `julia`, then `import Pluto; Pluto.run()`

- A browser tab opens; paste the notebook's path into *Open a notebook*
- Running the .jl file with `Shift+Enter` instead stops with "Package Plots not found": rule 3

An AI coding agent, in four words

An agent is a language model wrapped in a **loop** that can read your files, edit them and run commands.



- **Model**: no memory, no access to your disk. **Harness**: builds the context, runs the tools, shows you.
- **Tools**: the actions it may take. **Instructions**: an `AGENTS.md` that encodes your own rules.
- The model never touches your disk; the harness does, and Git shows you what it did

opencode: how we use it

Set-up. One install command; then `/connect` to pick a model provider, `/models` to pick a model

- **GitHub Copilot** gives several strong models, free with the **Student Developer Pack**
  - Apply here: `github.com/education/students`
- Free alternatives: OpenRouter's `:free` models, Google Gemini's free tier
- An API key is a password that spends money: never paste one into a file, a chat or a notebook

The supervised session you will run today, in your own repository with a clean `git status`:

1. **Plan mode** (press Tab until the footer says *Plan*): the agent proposes, changes nothing
2. Ask for something small and checkable: a function with a test you already know the answer to
3. Read the plan. Does it do **only** what you asked? Then Tab to *Build*
4. **Verify**: `git diff`, run it, read the output; only then `git commit`
5. `/init` writes an `AGENTS.md`; add one rule of your own

Using AI in this module

- **Allowed**, for the problem sets and in the labs, on one condition: **own it!**
  - You should be able to understand, explain and defend the code you wrote
- The problem sets assess **your** understanding of the models and methods, not the agent's
- Code you cannot explain when asked in a tutorial is treated as not your work
- The habits from the last two slides are how you stay on the right side of this

Now it is your turn

- Open the **setup guide** and work through it **in order**
  - `aseem-patel.github.io/EC991/setup/week01_tutorial.html`
- Every course file is in the course repository: `github.com/Aseem-Patel/EC991`

Part	You install or do	You know it worked when
1	Terminal, Git, GitHub account	`git -version` prints a version
2	Julia via juliaup; `add Pluto`	`1 + 1` in the REPL; `juliaup status`
3, 4	VS Code, Julia extension, a project	`Shift+Enter` runs a line; `Project.toml` appears
5	Smoke test `week01_check.jl`	Seven `PASS` lines
6	`ec991-work` repository	Your commit is visible on GitHub
7, 8	Pluto and the Julia-basics notebook	Exercises 2, 4 and 5 turn green
9	opencode, provider, supervised session	A reviewed diff, committed; `AGENTS.md` exists
11	Checklist	Every box ticked

- Stuck for more than a moment? Check the **troubleshooting table** in Part 11, then call me over
- Finished early? Sections 6 to 8 of the notebook, then the Julia and Python table in Part 8

The errors you will meet

Symptom	Cause and fix
`julia: command not found`	The terminal was open before the install. Close it and open a new one.
`Package Plots not found`, in a script	Wrong environment active. `]` `activate .` then `instantiate` or `add Plots`.
`Package Plots not found`, on the notebook	You ran the notebook as a script. Open it through Pluto.
Pluto: *Multiple definitions for x*	The same name is defined in two cells. Delete one.
Pluto: *Multiple expressions in one cell*	Wrap the cell in `begin ... end` or `let ... end`.
The first plot takes a long time	Normal: Julia compiles Plots on first use. The second plot is instant.
Git refuses to commit	Set `user.name` and `user.email` (Part 1).
opencode: *no provider configured*	Run `/connect` again; `opencode auth list` shows stored credentials.